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BE CSE
CSA 1603

CD4 ATL-1

1. Hospital patient segmentation

given data:

Patient	Age	monthly medical	Health Risk	Cluster	SSE
P_1	28	12 000	65	1	600
P_2	55	25 000	40	2	350
P 3	42	20, 000	75	3	220
P 4	30	11,000	70	4	200
P 5	60	28000	35	5	190

Interpretation:

The elbow method is used to determine the suitable number of cluster in K means. The SSE decreases as the number of cluster in K-mean.

K=1 → SSE = 600

K=2 → SSE = 350

K=3 → SSE = 220

K=4 → SSE = 200

K=5 → SSE = 190

Optimal no of cluster:

Optimal K = 3 cluster

The three clusters can represent different patient groups such as

* High risk patients :- Patients with high health risk dues

* High expenses patients: Patients requiring greater monthly medical expenditure

* Lower risk patients: Patients who many requires compare less intensive healthcare resources.

Effect healthcare Decision - making :-

selecting three cluster allows the healthcare organization to segment patients more meaningful stead of treating all patients equally. High risk patients can receive closer monitoring and presenting care, High expense patient can be studied for specialized treatest or cost management programs. Lower risk Patient can receive routine healthcare services. If K is too small important different better patient may be hidden. If K is too large, the segmentation becomes unnecessarily complicated and many